DiaTool-DPO: Multi-Turn DPO for Tool-Augmented LLMs

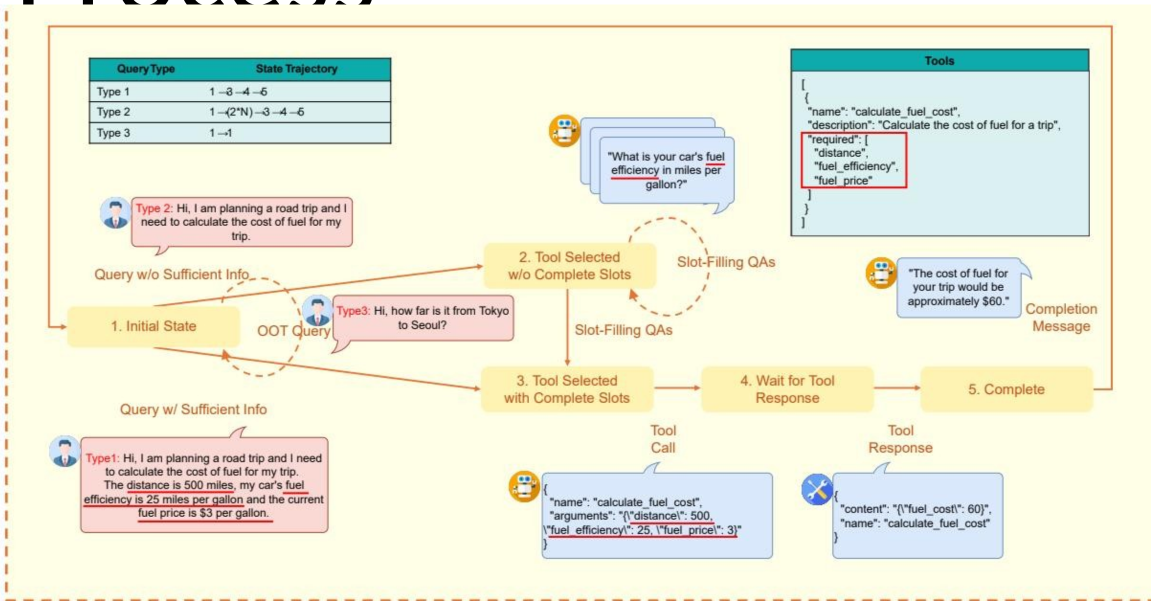
Enhancing Tool-Augmented LLM Dialogue via Direct Preference Optimization

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

- TA-LLMs trained only with Supervised Fine-Tuning (SFT) struggle with conversational flow:
  - Failure Modes:
    - Failing to ask follow-up questions when arguments are missing.
    - Making hallucinated tool invocations instead of gathering necessary data.
    - Failing to reject inappropriate tool calls when out-of-scope requests are made.
- Motivation: Direct Preference Optimization (DPO) is required to train robust dialogue capabilities by learning from paired (correct

Modeling TA-LLM Dialogue as a 5-State Markov Decision Process

1. Initial State: No history. Out-of-scope requests return a rejection message and remain here.
2. Tool Selected w/o Complete Slots: Tool is identified but required parameters are missing. Slot-filling QA occurs here.
3. Tool Selected w/ Complete Slots: Both tool selection and all argument values are determined.
4. Wait for Tool Response: Tool call executed, awaiting results.
5. Complete: Tool results received, completion message generated for the user.



Three Query Types Define Distinct State Trajectories

Query Type	State Trajectory	Description & Example
Type 1	1 → 3 → 4 → 5	Query contains all required arguments. No slot-filling needed.
Type 2	1 → (2*N) → 3 → 4 → 5	Query lacks some arguments. Slot-filling QA repeats N times until all required fields are gathered.
Type 3	1 → 1	Requested functionality is unavailable. Tool call must be rejected.

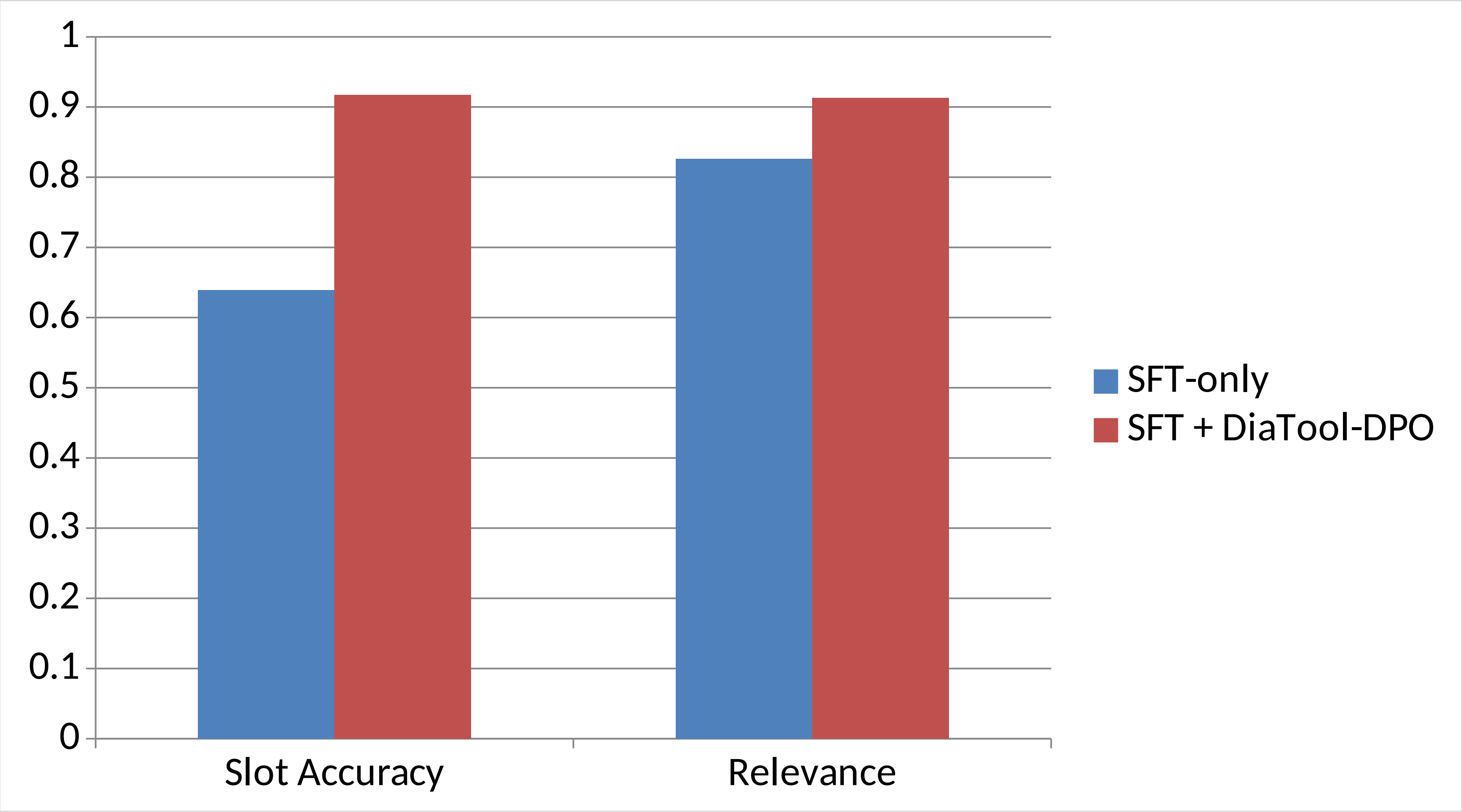
Automatic Paired Trajectory Construction Without Human

Query	Chosen traj.	Rejected traj.	Learning lesson	Count	Difficulty
Type 1	1→3→4→5	1→2→3→4→5	Prevent redundant slot-filling	2,089	Easy
Type 1	1→3→4→5	1→(2*N)→3→4→5	Prevent redundant slot-filling	562	Hard
Type 1	1→3→4→5	1→(2*M)→3→4→5	Prevent redundant slot-filling	2,530	Hard
Type 1	1→3→4→5	1→1	Tool call accept	2,090/562	Easy, Hard
Type 2	1→2→3→4→5	1→3→4→5	Prevent slot hallucination	2,089	Easy
Type 2	1→(2*N)→3→4→5	1→3→4→5	Prevent slot hallucination	562	Hard
Type 2	1→(2*N)→3→4→5	1→(2*M)→3→4→5	Prevent slot hallucination	2,530	Hard
Type 2	—	1→1	Tool call accept	2,089/562	Easy, Hard
Type 3	1→1	1→3→4	Tool call reject	567	Hard
Type 3	1→1	1→(2*N)→3→4	Tool call reject	562	Hard

SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

Model	Method	Call	Completion	Slot	Relevance	Micro Avg.	Macro Avg.
Prop.-8B	DiaTool-DPO-only	0.314	0.700	0.833	0.609	0.575	0.614
Prop.-8B	SFT-only	0.900	0.916	0.694	0.913	0.870	0.856
Prop.-8B	SFT + DiaTool-DPO	0.886	0.929	0.833	0.826	0.884	0.868
Prop.-3.1B	DiaTool-DPO-only	0.357	0.551	0.528	0.391	0.455	0.457
Prop.-3.1B	SFT-only	0.771	0.817	0.750	0.826	0.790	0.791
Prop.-3.1B	SFT + DiaTool-DPO	0.743	0.871	0.833	0.826	0.765	0.818
LLaMA-3-8B	DiaTool-DPO-only	0.029	0.449	0.056	0.261	0.205	0.199
LLaMA-3-8B	SFT-only	0.843	0.957	0.639	0.826	0.844	0.816
LLaMA-3-8B	SFT + DiaTool-DPO	0.857	0.929	0.917	0.913	0.905	0.904

LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relevance with DiaTool-DPO



Key Takeaways: RL-Based Dialogue Control for Production TA-LLMs

- MDP formulation enables systematic trajectory pairing without human annotation.
- Automatic dataset construction eliminates the need for expensive human labeling.
- DPO strongly complements SFT for dialogue control (SFT + DPO gives the best balanced performance).
- The approach is language-agnostic and robust across multiple model scales.
- Opens a path to GPT-4o-level conversational